Lecture 14 (B13): Large OLG Benchmark — 56 Cohorts (AGS 2022)

Bond market, lifecycle capital, training diagnostics

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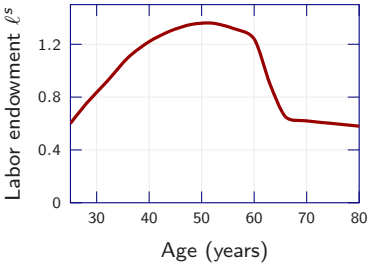
III.1 Setup — the Leading Example of AGS (2022)

Same recipe as Part II. What changes:

- Agents live for $A = 56$ periods (ages 25–80); one period = one year.
- CRRA utility $u(c) = c^{1-\gamma}/(1-\gamma)$ with $\gamma = 2$ (not log).
- Hump-shaped labor endowment ℓ^s (Kotlikoff-style; right panel).
- Persistent Markov shocks on (η, δ) (not i.i.d.).
- Two assets: capital k^s and bonds b^s.
- Two constraints: borrowing $k'^s \geq 0$ and collateral $k'^s + \kappa b'^s \geq 0$.
- Capital adjustment costs $\Psi^s = \frac{\zeta}{2}(k'^s - rk^s)^2$.

Stylised lifecycle labor (schematic)



Schematic shape only, rises through working years, drops sharply at retirement, low but nonzero after. See AGS (2022) for the exact ℓ^s used in the notebook.

III.2 Households — Preferences and Savings Technology

An agent of age s alive at t maximises

$$\mathbb{E}\left[\sum_{i=0}^{A-s} \beta^i \, u\big(c_{t+i}^{s+i}\big)\right], \qquad u(c) = \frac{c^{1-\gamma}}{1-\gamma}, \qquad \gamma = 2. \qquad (1')$$

Two saving instruments (instead of one):

- Capital k_{t+1}^{s+1}: yields gross return r_{t+1} next period; incurs an adjustment cost $\Psi_t^s = \frac{\zeta}{2}(k_{t+1}^{s+1} - r_t k_t^s)^2$.
- Bonds b_{t+1}^{s+1}: traded at price q_t today, pay one unit of consumption next period. Zero net supply: $\sum_{s=1}^{A} b_t^s = 0$.

Savings become next-period asset holdings ($\forall s \in \{1, \ldots, A-1\}$):

$$k_{t+1}^{s+1} \;=\; a_{t,k}^s, \qquad b_{t+1}^{s+1} \;=\; a_{t,b}^s. \qquad (2')$$

III.3 Households — Constraints, Budget, Boundary

Two constraints on the capital–bonds pair (k'^s, b'^s)**:**

$$k'^s \;\geq\; 0 \quad \text{(borrowing constraint)}, \qquad\qquad k'^s \;+\; \kappa\, b'^s \;\geq\; 0 \quad \text{(collateral constraint)}. \qquad (3')$$

Budget constraint of cohort s (bonds enter at price q_t on the left-hand side, pay off on the right):

$$c_t^s \;+\; k_{t+1}^{s+1} \;+\; q_t\, b_{t+1}^{s+1} \;+\; \Psi_t^s \;=\; r_t\, k_t^s \;+\; b_t^s \;+\; \ell_t^s\, w_t. \qquad (4')$$

Boundary conditions (born and die without assets):

$$k_t^1 = 0,\;\; b_t^1 = 0; \qquad\qquad k_{t+1}^{A+1} = 0,\;\; b_{t+1}^{A+1} = 0.$$

Bond parameter $\kappa \geq 0$ governs how much bond-short position capital collateralises; $\kappa = 0$ is a hard short-sale constraint on bonds.

III.4 Firms and Markets — Adding the Bond Market

Representative firm. Same Cobb–Douglas as Part II:

$$f(K, L, z) \;=\; \eta(z)\, K^{\alpha}\, L^{1-\alpha} \;+\; K\big(1 - \delta(z)\big).$$

Labor now aggregates over the hump-shaped profile: $L_t = \sum_{s=1}^{A} \ell^s$ (a constant).

Markets cleared in equilibrium.

1. Labor: $L_t = \sum_{s=1}^{A} \ell^s$.
2. Capital: $K_t = \sum_{s=1}^{A} k_t^s$.
3. Bonds (new): $0 \;=\; \displaystyle\sum_{s=1}^{A} b_t^s$. (Zero net supply.)

Aggregate shocks. $z_t \in \{1, \ldots, 4\}$ as before, but transition π^z is persistent (not i.i.d.): (η, δ) have their own Markov chains, tensor-producted.

III.5 Competitive Equilibrium — Definition

Definition 1' (competitive equilibrium, benchmark)

Given initial conditions z_0, $\{k_0^s, b_0^s\}_{s=1}^{A-1}$, a competitive equilibrium is a collection of choices $\left\{(c_t^s,\ k_t^{s+1},\ b_t^{s+1})_{s=1}^{A-1}\right\}_{t=0}^{\infty}$, firm demands $(K_t, L_t)_{t\geq 0}$, and prices $(r_t, w_t, q_t)_{t\geq 0}$, such that:

1. Households maximize $(1')$ subject to $(2')$–$(4')$.
2. Firm maximises profits, same closed-form FOCs as Part II.
3. All three markets clear every period: capital, labor, bonds.

Relative to Part II: one new price (q_t), one new market-clearing condition (bonds), one new asset class per cohort in the choice vector. Everything else is the same.

III.6 Equilibrium Conditions (Benchmark)

Firm FOCs. Same closed form as Part II.

Two Euler equations per cohort, for $s = 1, \ldots, A-1$:

$$\text{(capital)} \;\; u'(c_t^s) \cdot \big(1 + \zeta\,(k'^s - r_t k_t^s)\big) \;=\; \beta\,\mathbb{E}\big[u'(c_{t+1}^{s+1})\, r_{t+1}\big] \;+\; \lambda_{\mathrm{b}}^s \;+\; \lambda_{\mathrm{c}}^s,$$

$$\text{(bonds)} \;\; u'(c_t^s) \cdot q_t \;=\; \beta\,\mathbb{E}\big[u'(c_{t+1}^{s+1})\big] \;+\; \kappa\,\lambda_{\mathrm{c}}^s.$$

Two KKT triples per cohort.

$$\lambda_{\mathrm{b}}^s \geq 0, \quad k'^s \geq 0, \quad \lambda_{\mathrm{b}}^s \cdot k'^s = 0; \qquad \lambda_{\mathrm{c}}^s \geq 0, \quad k'^s + \kappa b'^s \geq 0, \quad \lambda_{\mathrm{c}}^s \cdot (k'^s + \kappa b'^s) = 0.$$

Terminal cohort. Age A: consume everything.

So per state we have $4(A-1)$ per-agent residuals + 1 bond market-clearing residual = $\mathbf{4\,(A-1)+1}$ equations.

III.7 The DEQN Cost Function (Benchmark)

Five loss components, one cost function

$$\mathcal{L}_{D_{\text{train}}}(\rho) \;=\; \frac{1}{|D_{\text{train}}|}\,\frac{1}{4(A-1)+1}\sum_{x_j\in D_{\text{train}}}\left[\sum_{s=1}^{A-1} R^{s}(x_j)^2 \;+\; \left(e_{\text{MC},b}(x_j)\right)^2\right],$$

where $R^{s}(x_j)^2 \;:=\; \left(e^{s}_{\text{REE},k}\right)^2 + \left(e^{s}_{\text{REE},b}\right)^2 + \left(e^{s}_{\text{KKT},b}\right)^2 + \left(e^{s}_{\text{KKT},c}\right)^2.$

Component map.

- $e^{s}_{\text{REE},k}$, $e^{s}_{\text{REE},b}$: relative residuals of the capital and bond Euler equations.
- $e^{s}_{\text{KKT},b} = \lambda^{s}_{\text{b}} \cdot k'^{s}$: borrowing KKT.
- $e^{s}_{\text{KKT},c} = \lambda^{s}_{\text{c}} \cdot (k'^{s} + \kappa b'^{s})$: collateral KKT.
- $e_{\text{MC},b} = \sum_{s=1}^{A} b_t^{s}$: bond market clearing.

With $A = 56$ this is $4 \times 55 + 1 = 221$ loss terms per training state. The structure is the same as Part II: squared equilibrium residuals.

III.8 Functional Rational Expectations Equilibrium

FREE for the benchmark

$$f : \{1, \ldots, 4\} \times \mathbb{R}^A \times \mathbb{R}^A \longrightarrow \mathbb{R}^{4(A-1)+1}$$

$$f\left(\begin{bmatrix} z_t \\ \boldsymbol{k}_t \\ \boldsymbol{b}_t \end{bmatrix}\right) = \begin{bmatrix} \hat{k}_t'^1, \ldots, \hat{k}_t'^{A-1} \\ \hat{\lambda}_{b,t}^1, \ldots, \hat{\lambda}_{b,t}^{A-1} \\ \hat{q}_t^1, \ldots, \hat{q}_t^{A-1} \\ \hat{\mu}_t^1, \ldots, \hat{\mu}_t^{A-1} \\ \hat{p}_t \end{bmatrix}^\top$$

Output blocks: capital savings; borrowing multipliers; collateral $\hat{q}^h \equiv \hat{k}'^h + \kappa\, \hat{b}'^h$; collateral multipliers; equilibrium bond price (one scalar).

Such that $G\big([z_t, \boldsymbol{k}_t, \boldsymbol{b}_t],\ f([z_t, \boldsymbol{k}_t, \boldsymbol{b}_t])\big) = \mathbf{0}$ where G stacks the $4(A-1)+1$ equations of III.6.

With $A = 56$: minimal state dim $= 113$; the notebook input is 236 after engineered aggregates and transition probabilities. Output dim $= 4 \times 55 + 1 = \mathbf{221}$. Bond holdings are recovered as $\hat{b}'^h = (\hat{q}^h - \hat{k}'^h)/\kappa$, and softplus enforces $\hat{k}'^h, \hat{\lambda}_b^h, \hat{q}^h, \hat{\mu}^h \geq 0$.

III.9 Architecture — Benchmark

$$\mathcal{N}_\rho : \mathbb{R}^{236} \to \mathbb{R}^{221}$$

Sizes from notebook 08

Notebook input	236 (minimal state: 113)
Hidden layers	2×128 (classroom)
	2×1000 (production)
Output dim	$4(A-1) + 1 = 221$
Output layer	softplus on every output
Optimizer	Adam, lr 10^{-5}

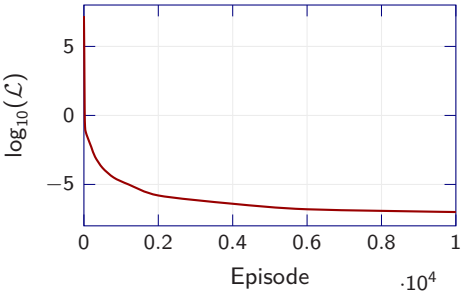
Output slicing

$$\mathcal{N}_\rho(x_t) = \left(\hat{k}'^{1:A-1},\ \hat{\lambda}_b^{1:A-1},\ \hat{q}^{1:A-1},\ \hat{\mu}^{1:A-1},\ \hat{p}\right).$$

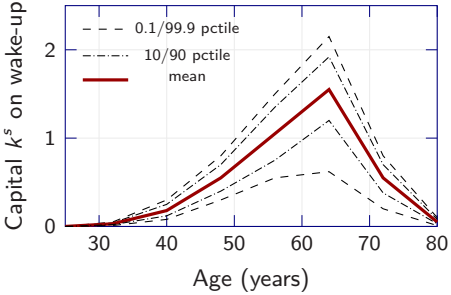
from a MLP (template) as Part II, scaled up and augmented by a bond-price head. Bond holdings are recovered

III.10 Training — Loss and Lifecycle Capital

Cost per episode (benchmark, 56 cohorts)



Lifecycle capital



Reading the plots

Left: loss falls sharply but oscillates because each episode re-simulates the cloud.
Right: lifecycle capital is hump-shaped: accumulation during working years, decumulation after retirement. Bands are simulated percentiles.

III.11 Consumption Profiles by Aggregate State

What the notebook will plot

For each aggregate state $z \in \{1, 2, 3, 4\}$ we obtain a lifecycle consumption profile $c^s(z)$ over ages $s = 25, \ldots, 80$, evaluated on a long ergodic simulation from notebook 08.

The four aggregate states

z	$\xi(z)$	$\delta(z)$	expected consumption ranking
2	1.022	0.08	highest – best technology, low depreciation
4	1.022	0.11	high
1	0.978	0.08	low
3	0.978	0.11	lowest – bad technology, high depreciation

What to look for in the live plot

Four distinct profiles, ordered as in the table above. The exact magnitudes and crossover ages are produced by section 11 of the notebook.

III.12 Reading the Diagnostics

How do we know the network actually solved the equilibrium?

1. Loss level. $\log_{10} \mathcal{L} \lesssim -5$ means equilibrium residuals at the 10^{-3} level or smaller – AGS (2022) threshold.
2. Per-equation Euler errors. Plot mean $|e^s_{\mathrm{REE},k}|$ and $|e^s_{\mathrm{REE},b}|$ by age s; expect $\sim 10^{-3}$ everywhere.
3. Bond market clearing. $|e_{\mathrm{MC},b}| < 10^{-4}$ on the ergodic simulation path.
4. Borrow-binding fraction. Ages where $\lambda^s_{\mathrm{b}} > 0$; standard calibration: only the youngest few cohorts.
5. Ergodic simulation. Simulate 10 000 periods; plot lifecycle consumption + capital with percentile bands.

Hands-on: `08_OLG_Benchmark_DEQN.ipynb`. §1–5 match III.1–III.5; §6 = cost function (III.7); §7–8 = training (III.9); §9–11 = plots and diagnostics of III.10–III.12. Classroom runtime (128-wide, 2 000 episodes): $\approx$ 30 min on CPU.

Summary

What we accomplished

1. Introduced OLG as a finite-dimensional family of heterogeneous-agent models – well suited to DEQNs because the aggregate state stays finite-dimensional.
2. Applied the DEQN recipe twice:
   - Part II: the 6-agent Krueger–Kübler (2004) analytic model, validated against the closed form ($\lesssim 10^{-3}$ mean Euler error).
   - Part III: the 56-cohort leading-example benchmark of AGS (2022): CRRA, hump labor, persistent shocks, two assets, two borrowing constraints, capital adjustment costs.
3. Same 9-step recipe both times: setup, households I/II, firms, equilibrium, conditions, cost, FREE, architecture.

One line to remember

Writing down a DEQN = writing down the equilibrium conditions G. Everything else (network, SGD, sampling) is boilerplate.

References and Further Reading

- Azinovic, M., Gaegauf, L., & Scheidegger, S. (2022). *Deep equilibrium nets.* International Economic Review 63(4), 1471–1525. [notebook 08 implements §3 of this paper].
- Krueger, D., & Kübler, F. (2004). *Computing equilibrium in OLG models with stochastic production.* Journal of Economic Dynamics & Control 28(7), 1411–1436. [notebook 07 implements their 6-agent analytic model].
- Fischer, A. (1992). *A special Newton-type optimization method.* Optimization 24, 269–284. [FB-complementarity function].
- Code: `https://github.com/sischei/DeepEquilibriumNets`.

Bridge to Toolkit T1: Agentic Programming

Where the technical content rests. At this point we have a complete discrete-time DEQN toolkit: feed-forward policy net + softplus output, Euler/KKT residuals + Fischer–Burmeister, simulation-based sampling, NAS-tuned architecture, and ReLoBRaLo-balanced losses, all of it scaling from a single-shock RBC sandbox up to a 56-cohort OLG benchmark with two binding constraints.

The agentic-programming toolkit (no lecture script chapter). We turn the workflow on its head and treat the model file, the loss, and the diagnostics as artefacts that an LLM-driven coding agent can author, refactor, and debug in conversation. Concretely, you will:

- Use Claude Code / Cursor / Codex-style agents as pair programmers on a fresh DEQN model derived from the chapters of Days 2–4.
- Practice the loop specification → scaffold → run → diagnose → refactor, with the agent doing the typing and you driving the economics.
- Connect the slide-deck → notebook → figure → commit pipeline that the manuscript itself was built with.

Self-study extension (notebooks 10–12, deck 08). Heterogeneous agents on a histogram (Young's method), continuum-of-agents DEQN, and Krusell–Smith with deep learning. These pieces sit *above* today's OLG content and can be revisited any time before the continuous-time PINN material.